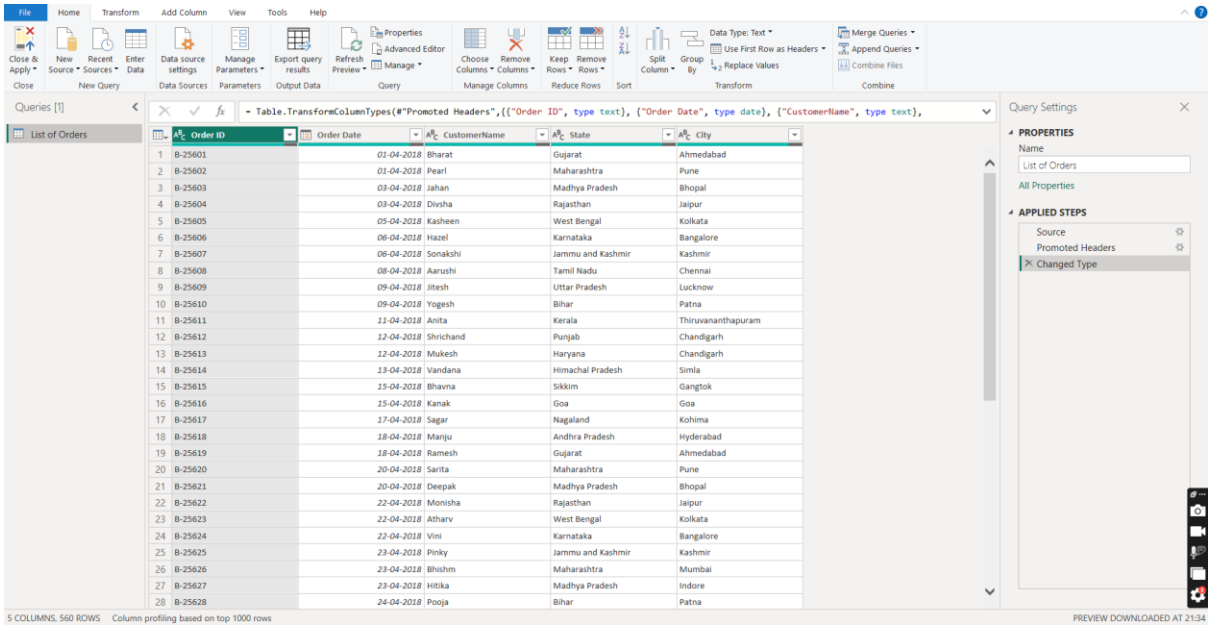
1. Data Import:

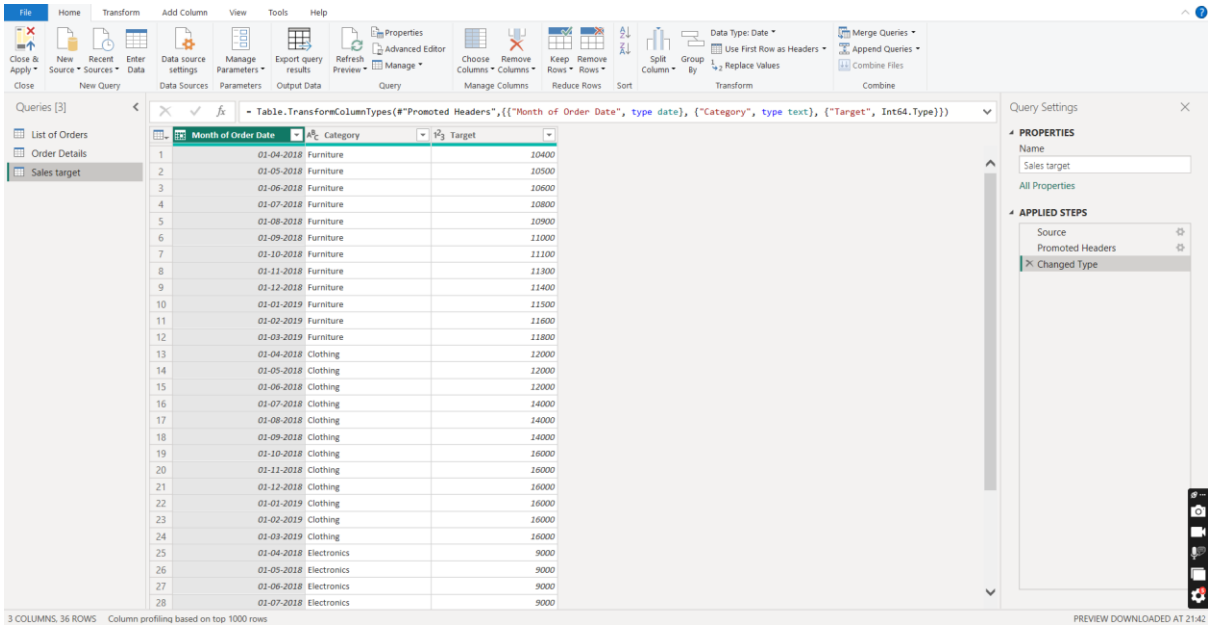
○ Import List of Orders.csv into Power BI.

○ Open the list of orders in the Power Query Editor using the Transform Data option.



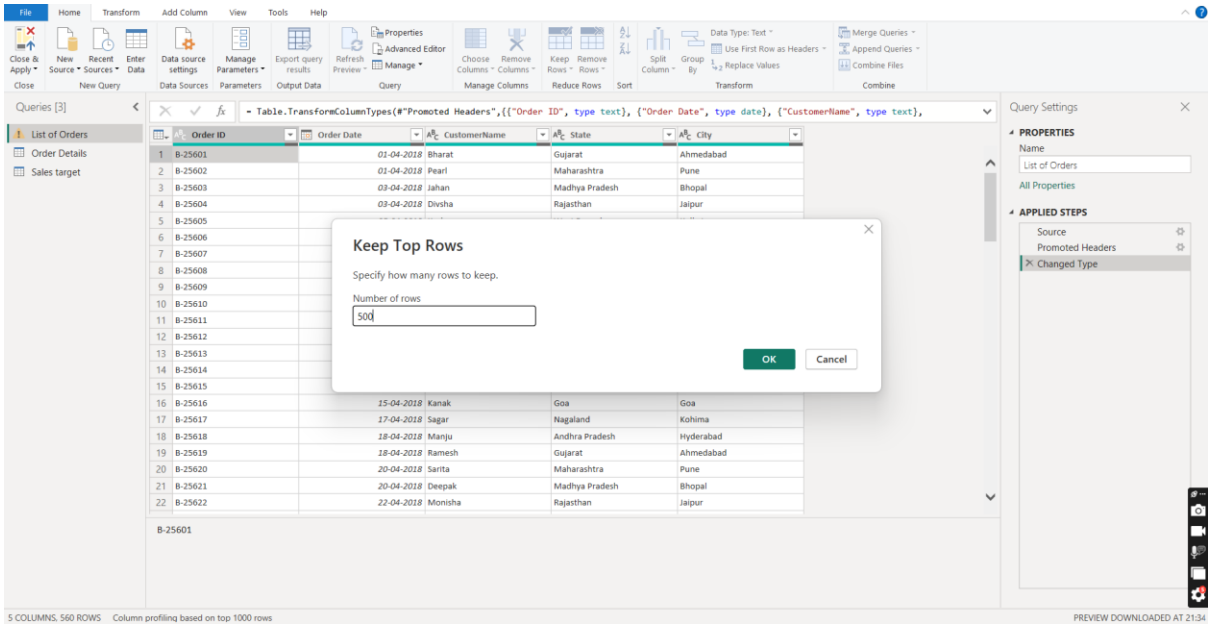
	Order ID	Order Date	CustomerName	State	City
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	09-04-2018	Yogesh	Bihar	Patna
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
13	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
15	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
16	B-25616	15-04-2018	Kanak	Goa	Goa
17	B-25617	17-04-2018	Sagar	Nagaland	Kohima
18	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19	B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
20	B-25620	20-04-2018	Sarita	Maharashtra	Pune
21	B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
22	B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
23	B-25623	22-04-2018	Atharv	West Bengal	Kolkata
24	B-25624	22-04-2018	Vini	Karnataka	Bangalore
25	B-25625	23-04-2018	Pinky	Jammu and Kashmir	Kashmir
26	B-25626	23-04-2018	Bhishm	Maharashtra	Mumbai
27	B-25627	23-04-2018	Hitika	Madhya Pradesh	Indore
28	B-25628	24-04-2018	Pooja	Bihar	Patna

○ Import Order Details.csv and Sales Target.csv into Power Query Editor.

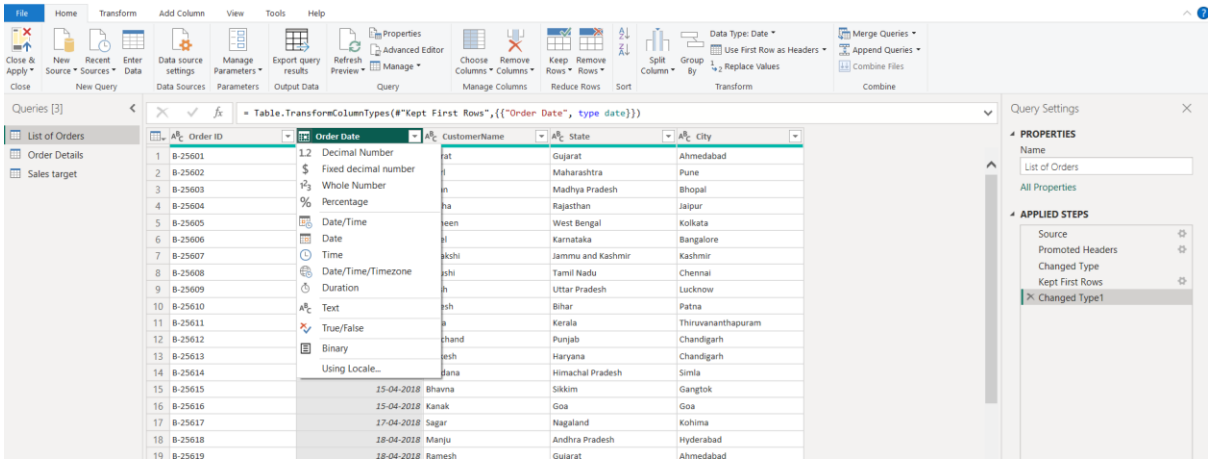


	Month of Order Date	Category	Target
1	01-04-2018	Furniture	10400
2	01-05-2018	Furniture	10500
3	01-06-2018	Furniture	10600
4	01-07-2018	Furniture	10800
5	01-08-2018	Furniture	10900
6	01-09-2018	Furniture	11000
7	01-10-2018	Furniture	11100
8	01-11-2018	Furniture	11300
9	01-12-2018	Furniture	11400
10	01-01-2019	Furniture	11500
11	01-02-2019	Furniture	11600
12	01-03-2019	Furniture	11800
13	01-04-2018	Clothing	12000
14	01-05-2018	Clothing	12000
15	01-06-2018	Clothing	12000
16	01-07-2018	Clothing	14000
17	01-08-2018	Clothing	14000
18	01-09-2018	Clothing	14000
19	01-10-2018	Clothing	16000
20	01-11-2018	Clothing	16000
21	01-12-2018	Clothing	16000
22	01-01-2019	Clothing	16000
23	01-02-2019	Clothing	16000
24	01-03-2019	Clothing	16000
25	01-04-2018	Electronics	9000
26	01-05-2018	Electronics	9000
27	01-06-2018	Electronics	9000
28	01-07-2018	Electronics	9000

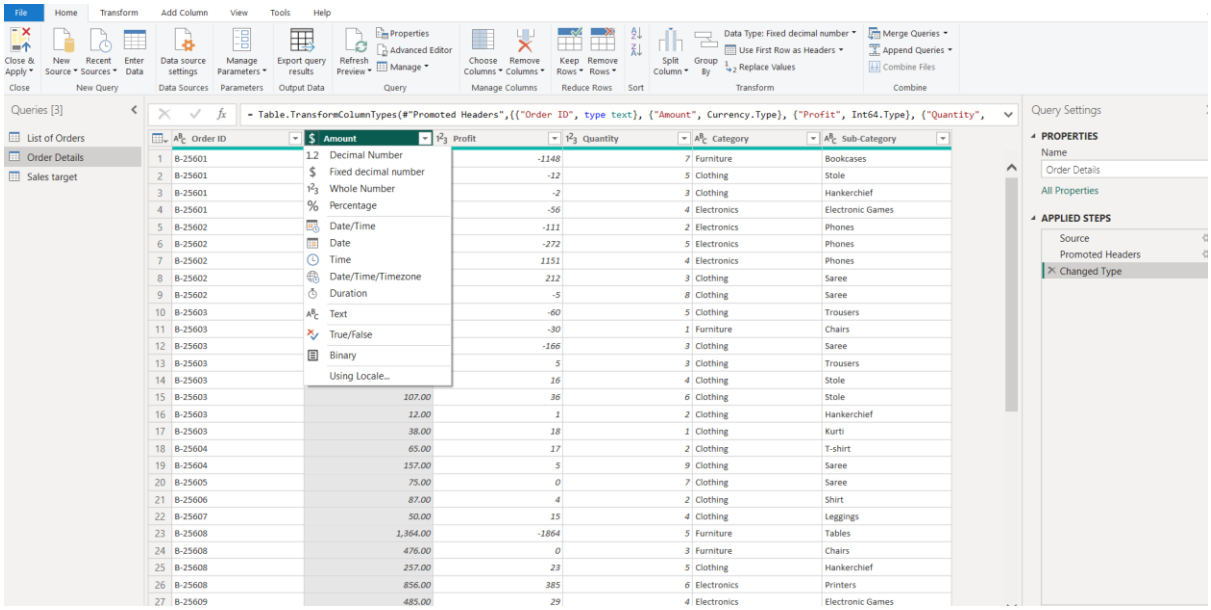
2.1 Restrict the List of Orders table to the first 500 rows.

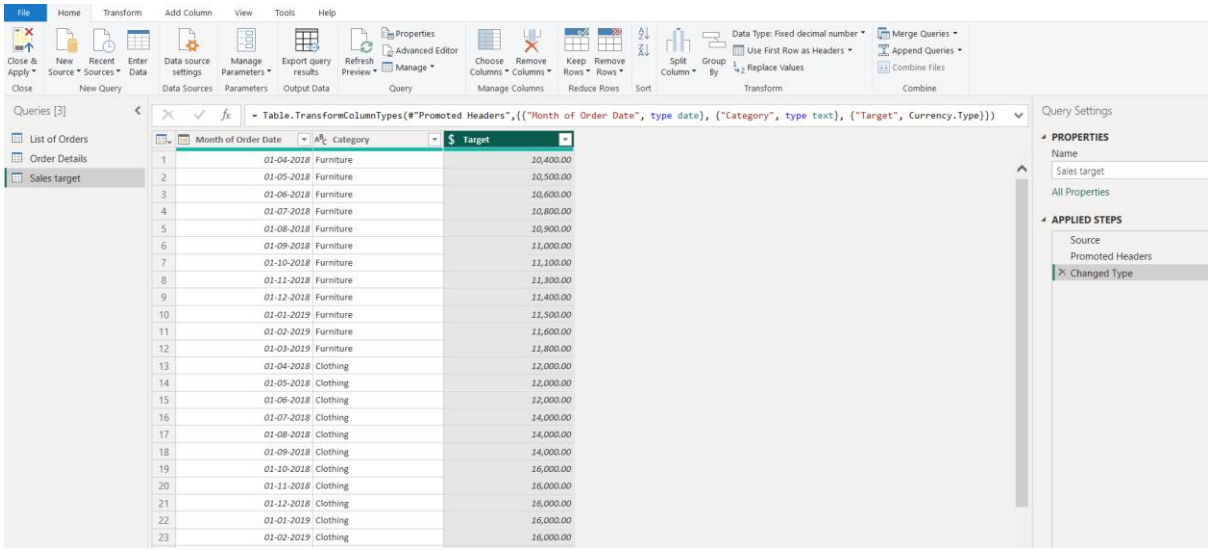


2.2 Set the Order Date column to the Date data type.

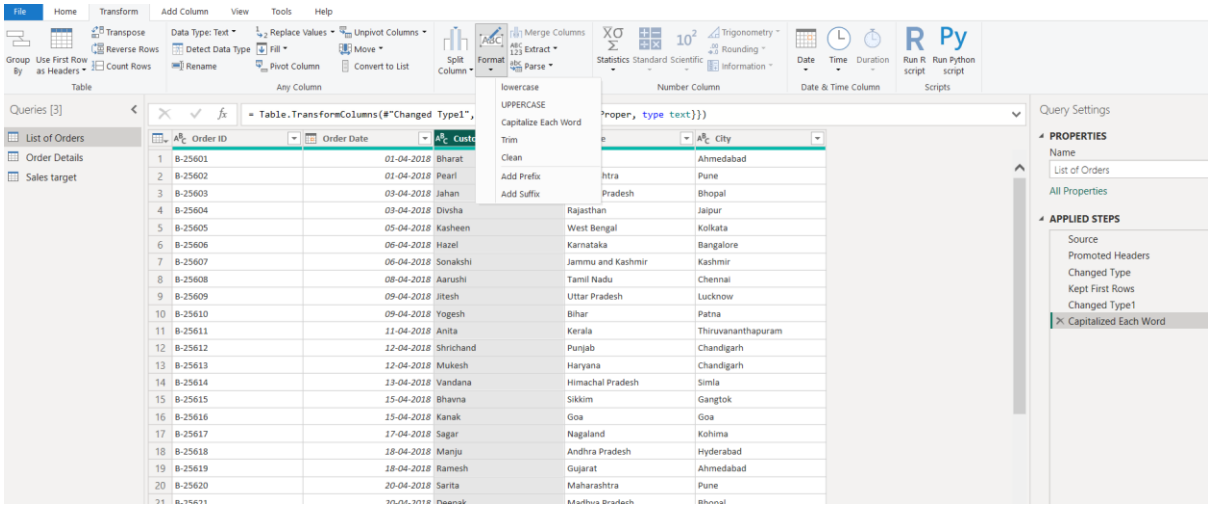


2.3 Change the Amount and Target columns to fixed decimal numbers.





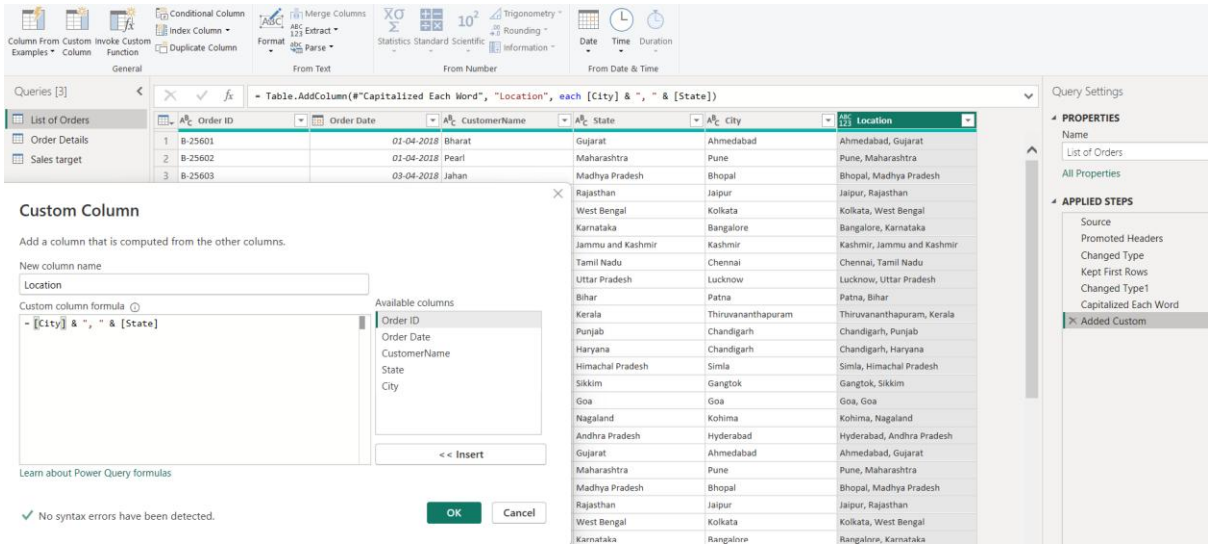
3.1 Format the Customer Name column to proper case to ensure consistent capitalization.



	Order ID	Order Date	Customer Name	State	City
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	09-04-2018	Yogesh	Bihar	Patna
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
13	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
15	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
16	B-25616	15-04-2018	Kanak	Goa	Goa
17	B-25617	17-04-2018	Sagar	Nagaland	Kohima
18	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19	B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
20	B-25620	20-04-2018	Sarita	Maharashtra	Pune

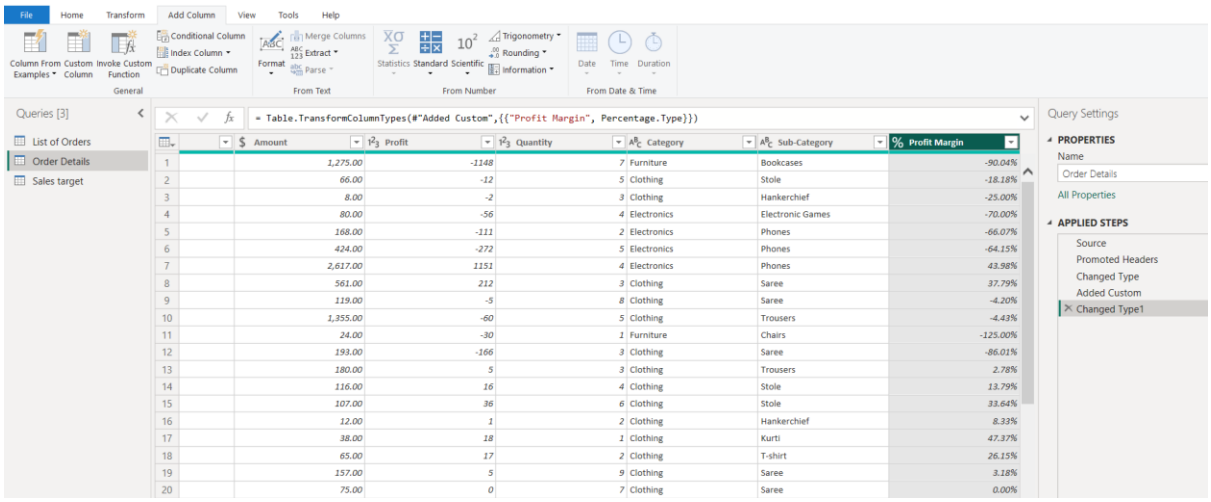
4.1 Merge the City and State columns to create a new column named Location in the format:

City, State



	Order ID	Order Date	CustomerName	State	City	Location
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune	Pune, Maharashtra
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal, Madhya Pradesh
				Rajasthan	Jaipur	Jaipur, Rajasthan
				West Bengal	Kolkata	Kolkata, West Bengal
				Karnataka	Bangalore	Bangalore, Karnataka
				Jammu and Kashmir	Kashmir	Kashmir, Jammu and Kashmir
				Tamil Nadu	Chennai	Chennai, Tamil Nadu
				Uttar Pradesh	Lucknow	Lucknow, Uttar Pradesh
				Bihar	Patna	Patna, Bihar
				Kerala	Thiruvananthapuram	Thiruvananthapuram, Kerala
				Punjab	Chandigarh	Chandigarh, Punjab
				Haryana	Chandigarh	Chandigarh, Haryana
				Himachal Pradesh	Simla	Simla, Himachal Pradesh
				Sikkim	Gangtok	Gangtok, Sikkim
				Goa	Goa	Goa, Goa
				Nagaland	Kohima	Kohima, Nagaland
				Andhra Pradesh	Hyderabad	Hyderabad, Andhra Pradesh
				Gujarat	Ahmedabad	Ahmedabad, Gujarat
				Maharashtra	Pune	Pune, Maharashtra
				Madhya Pradesh	Bhopal	Bhopal, Madhya Pradesh
				Rajasthan	Jaipur	Jaipur, Rajasthan
				West Bengal	Kolkata	Kolkata, West Bengal
				Karnataka	Bangalore	Bangalore, Karnataka

4.2 Create a custom column named Profit Margin calculated as: Profit / Amount (expressed as a percentage)



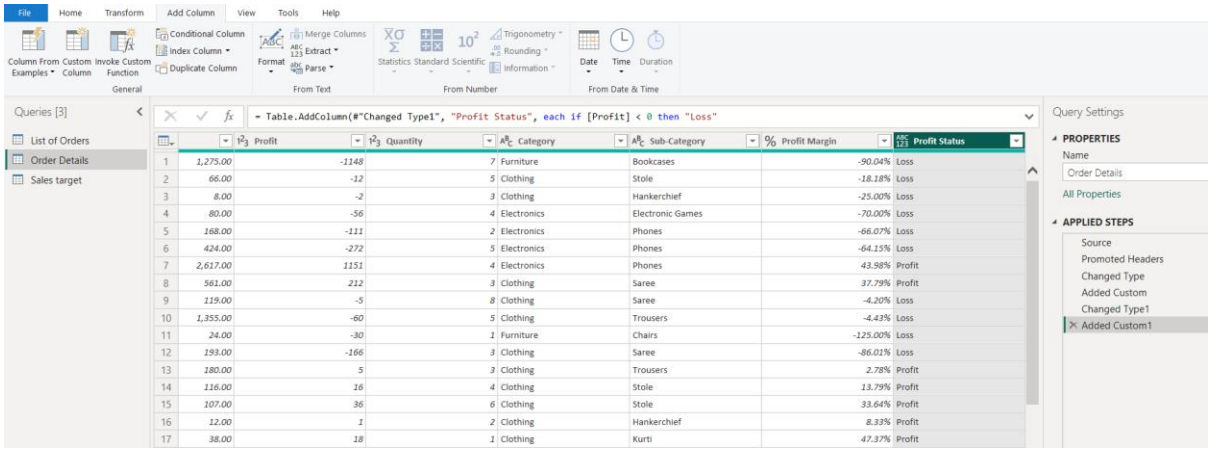
	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin
1	1,275.00	-1148	7	Furniture	Bookcases	-90.04%
2	66.00	-12	5	Clothing	Stole	-18.18%
3	8.00	-2	3	Clothing	Hankerchief	-25.00%
4	80.00	-56	4	Electronics	Electronic Games	-70.00%
5	168.00	-111	2	Electronics	Phones	-66.07%
6	424.00	-272	5	Electronics	Phones	-64.15%
7	2,617.00	1151	4	Electronics	Phones	43.98%
8	561.00	212	3	Clothing	Saree	37.79%
9	119.00	-5	8	Clothing	Saree	-4.20%
10	1,355.00	-60	5	Clothing	Trousers	-4.43%
11	24.00	-30	1	Furniture	Chairs	-125.00%
12	193.00	-166	3	Clothing	Saree	-86.01%
13	180.00	5	3	Clothing	Trousers	2.78%
14	116.00	16	4	Clothing	Stole	13.79%
15	107.00	36	6	Clothing	Stole	33.64%
16	12.00	1	2	Clothing	Hankerchief	8.33%
17	38.00	18	1	Clothing	Kurti	47.37%
18	65.00	17	2	Clothing	T-shirt	26.15%
19	157.00	5	9	Clothing	Saree	3.18%
20	75.00	0	7	Clothing	Saree	0.00%

5.1 Add a conditional column named Profit Status based on:

■ Profit < 0 → Loss

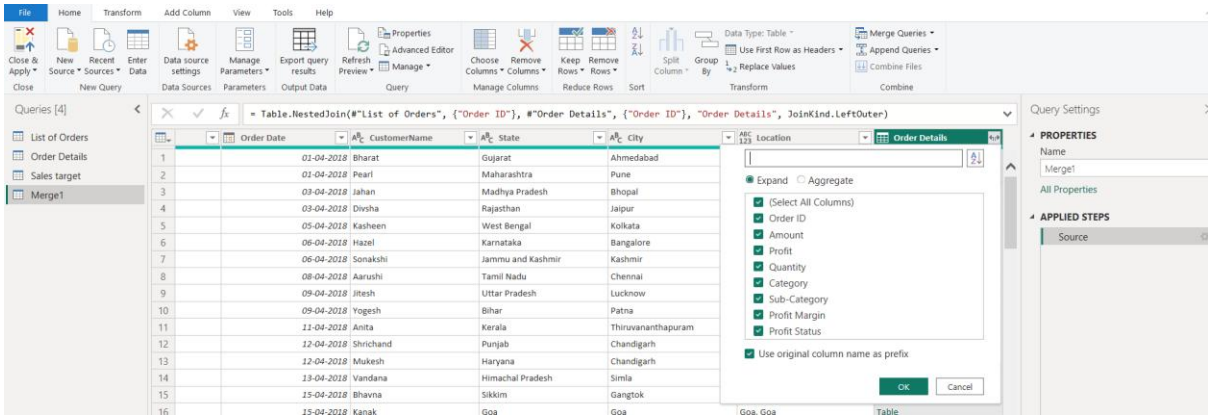
■ Profit = 0 → Break-Even

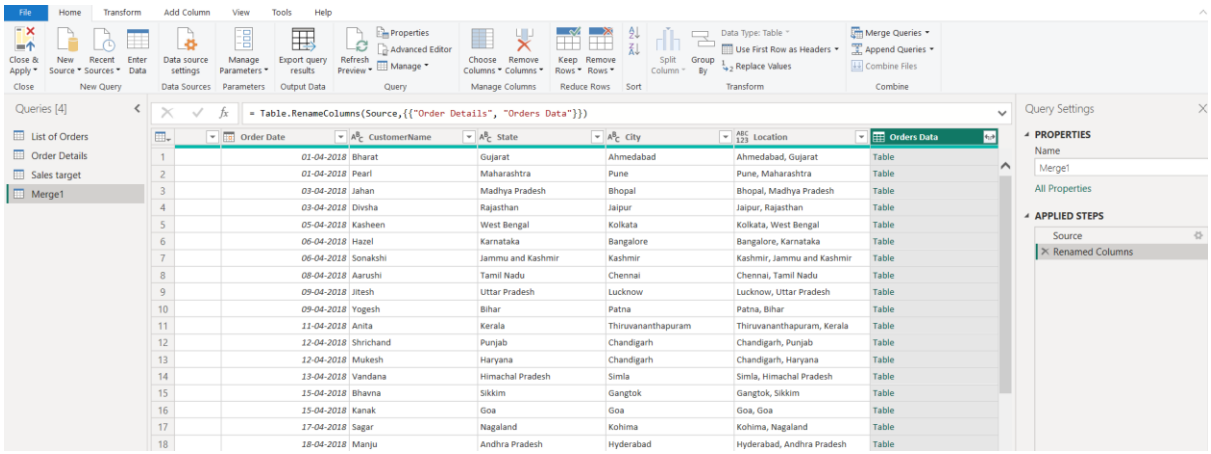
■ Profit > 0 → Profit



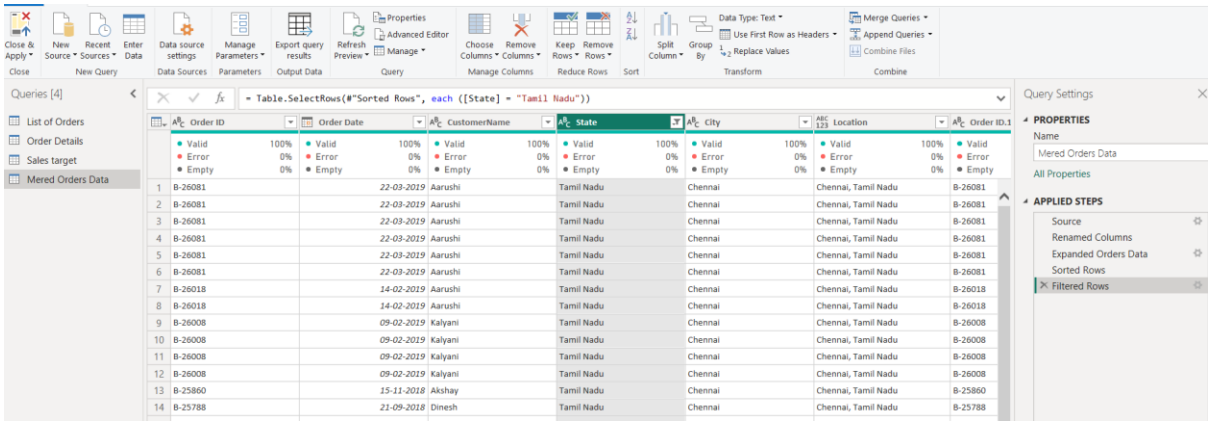
	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
1	1,275.00	-1148	7	Furniture	Bookcases	-90.04%	Loss
2	66.00	-12	5	Clothing	Stole	-18.18%	Loss
3	8.00	-2	3	Clothing	Hankerchief	-25.00%	Loss
4	80.00	-56	4	Electronics	Electronic Games	-70.00%	Loss
5	168.00	-111	2	Electronics	Phones	-66.07%	Loss
6	424.00	-272	5	Electronics	Phones	-64.15%	Loss
7	2,617.00	1151	4	Electronics	Phones	43.98%	Profit
8	561.00	212	3	Clothing	Saree	37.79%	Profit
9	119.00	-5	8	Clothing	Saree	-4.20%	Loss
10	1,355.00	-60	5	Clothing	Trousers	-4.43%	Loss
11	24.00	-30	1	Furniture	Chairs	-125.00%	Loss
12	193.00	-166	3	Clothing	Saree	-86.01%	Loss
13	180.00	5	3	Clothing	Trousers	2.78%	Profit
14	116.00	16	4	Clothing	Stole	13.79%	Profit
15	107.00	36	6	Clothing	Stole	33.64%	Profit
16	12.00	1	2	Clothing	Hankerchief	8.33%	Profit
17	38.00	18	1	Clothing	Kurti	47.37%	Profit

6.1 Merge the List of Orders and Order Details tables using Order ID.

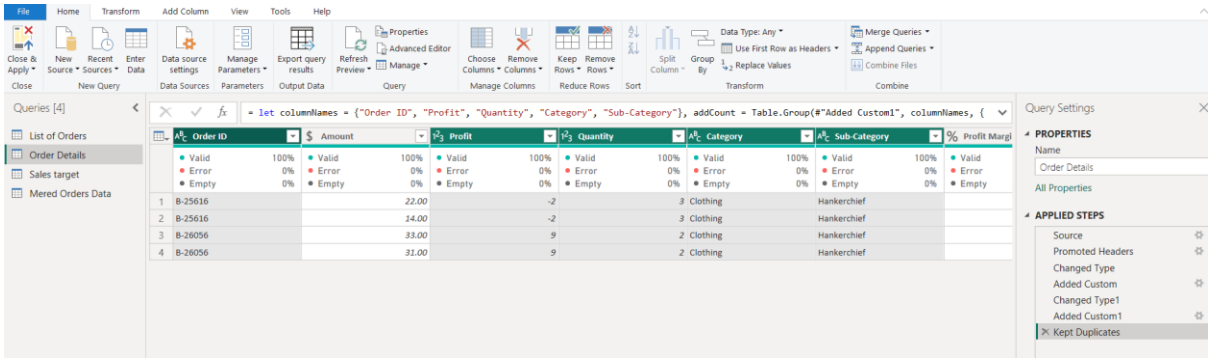




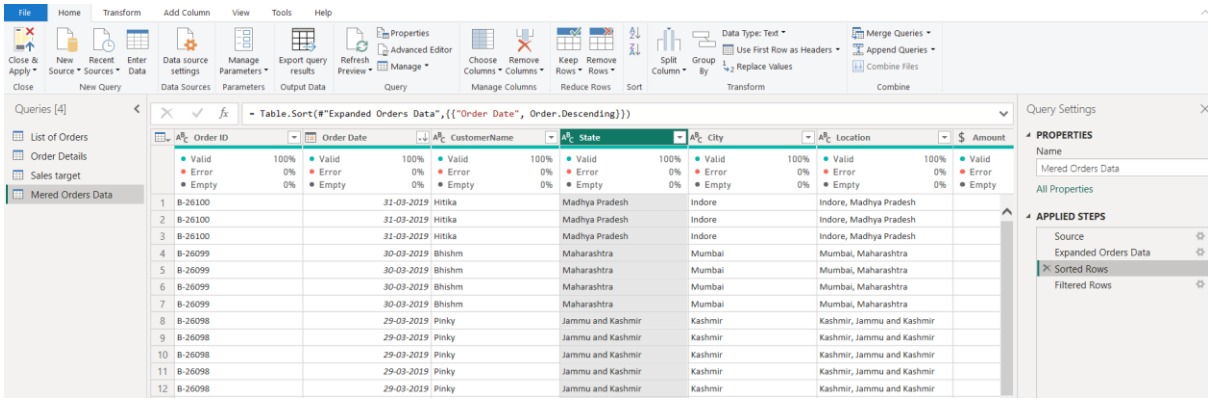
6.2 Name the merged output table as Orders Data.



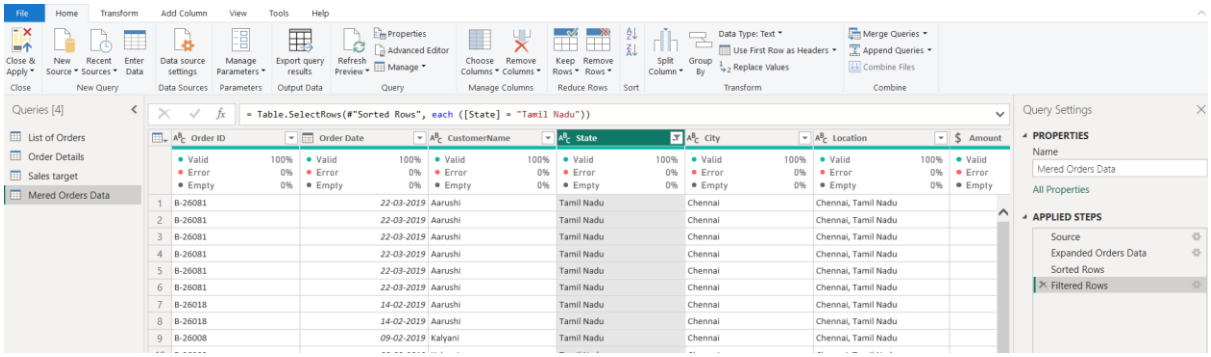
7.1 Identify missing values across all tables.



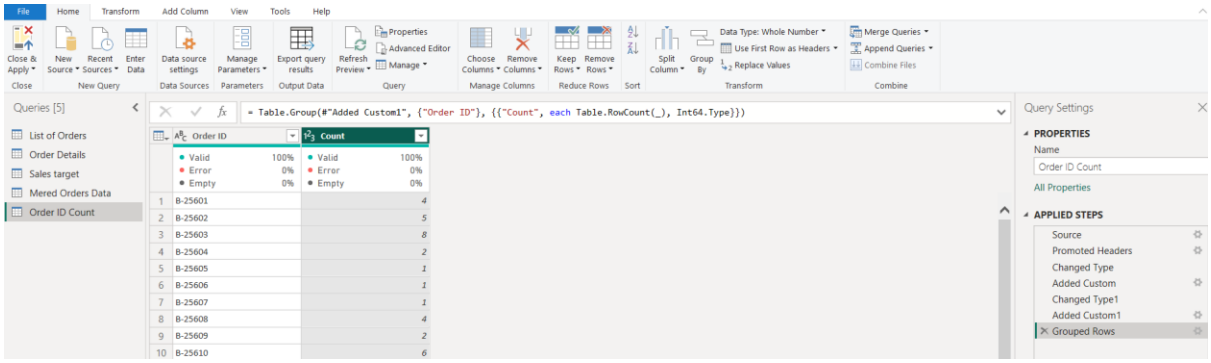
8.1 Sort records in the Orders Data table by Order Date in descending order.



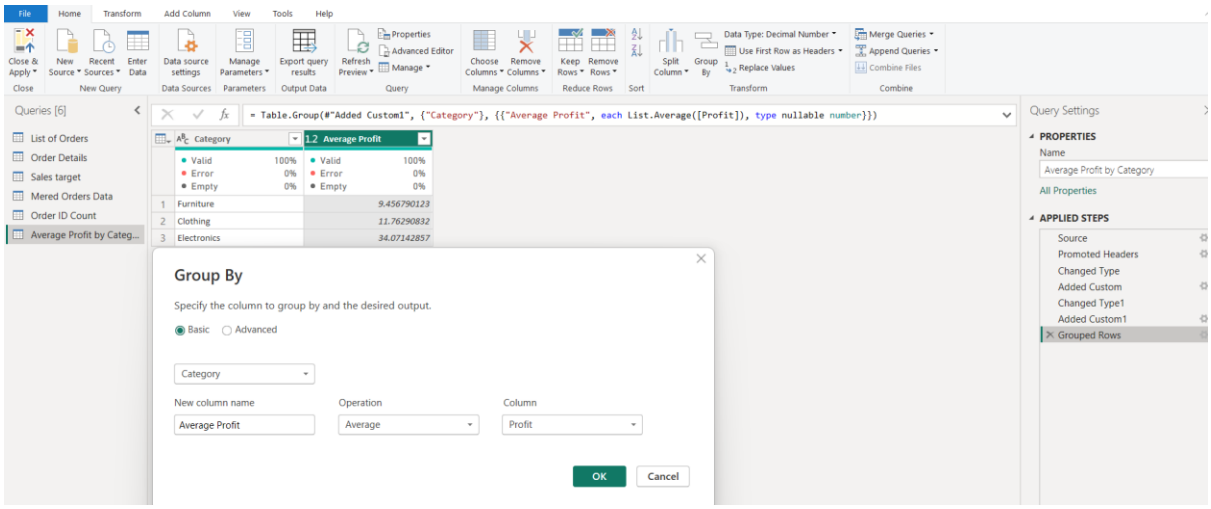
8.2 Apply filters to analyze orders from a specific state (e.g., Tamil Nadu).



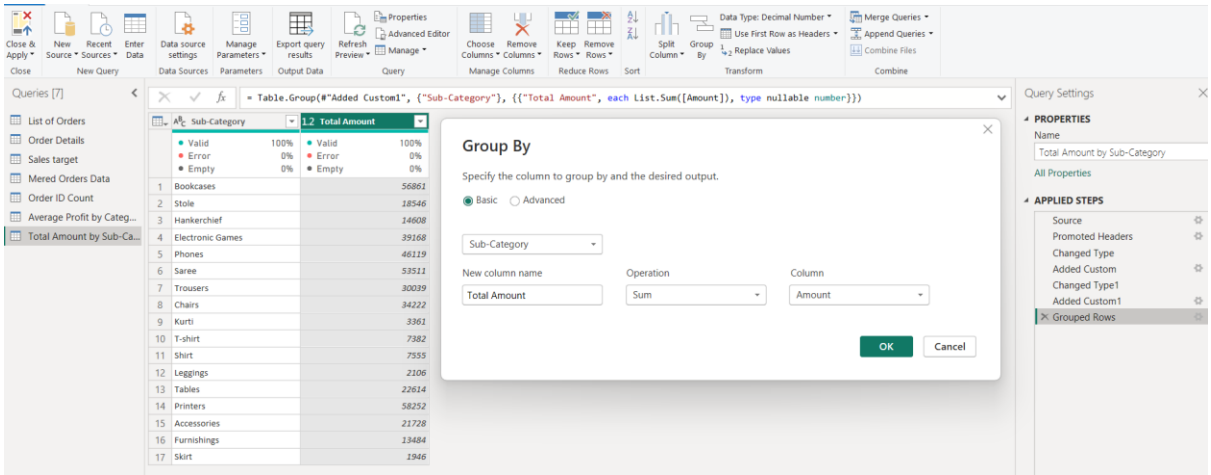
9.1 Count of Order ID



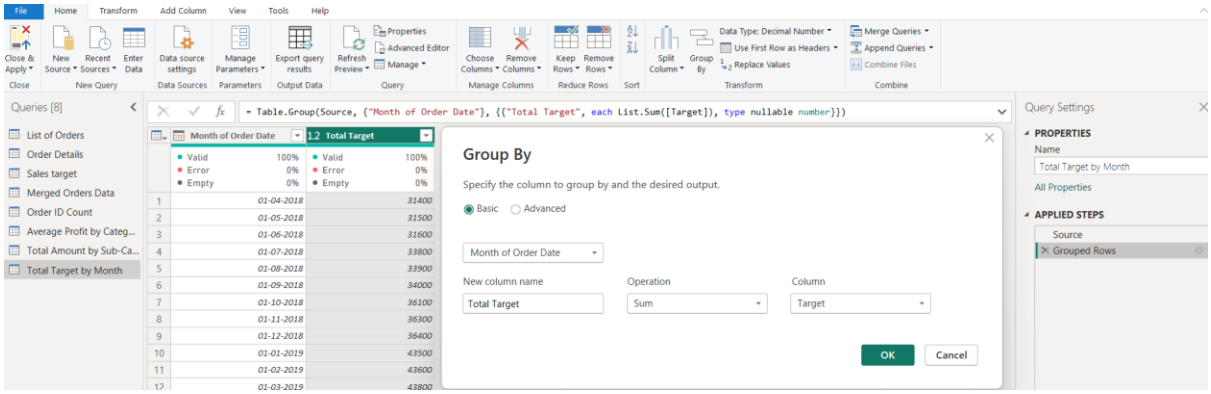
9.2 Average Profit by Category



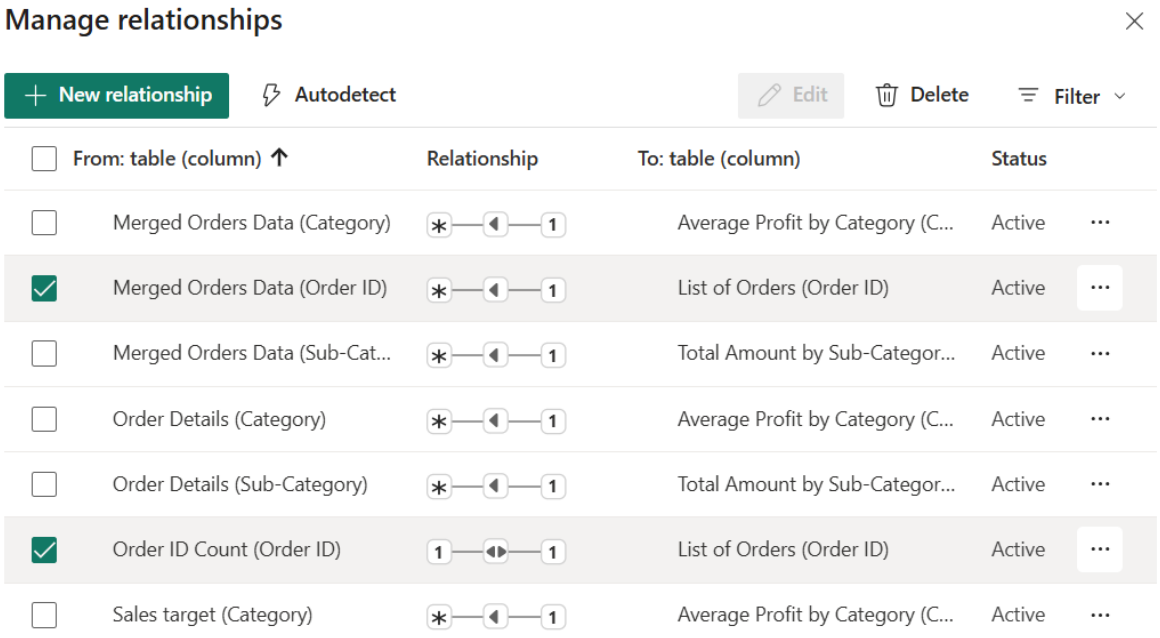
9.3 Total Amount by Sub-Category

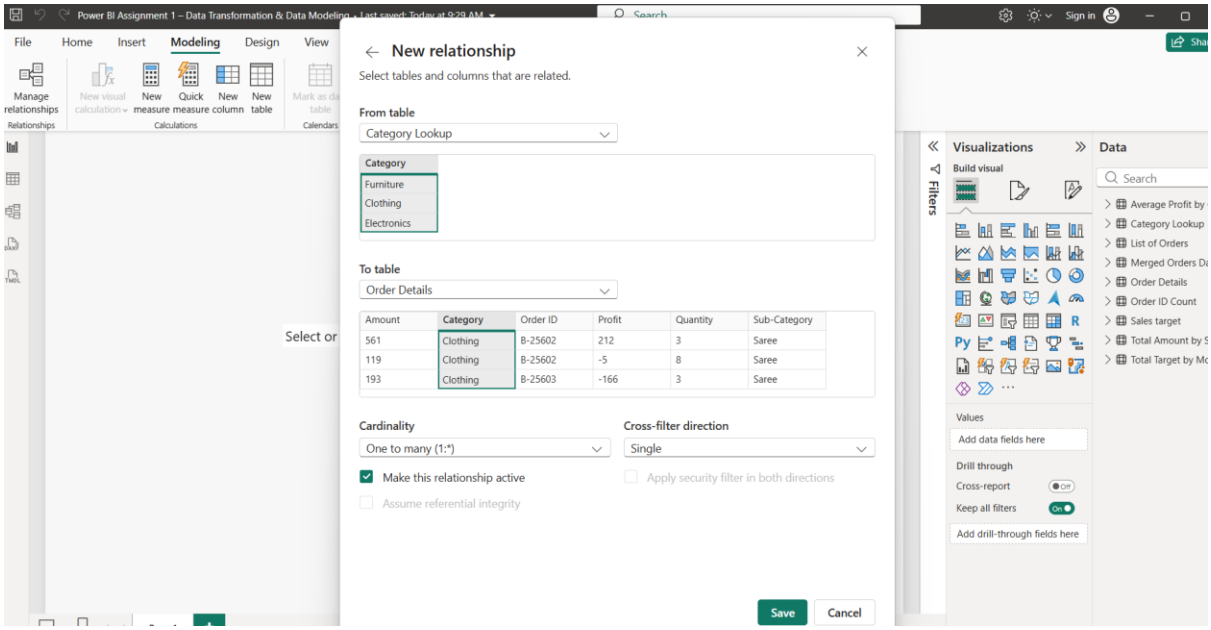


9.4 Duplicate the Sales Target table and aggregate the total Target by Month of order date.



10.1 List of Orders and Order Details using Order ID




Power BI Assignment 1 – Data Transformation & Data Modeling
File
Home
Insert
Modeling
Design
View
Manage relationships
New measure
Quick measure
New column
New table
Relationships
Calculations
New relationship
Select tables and columns that are related.
From table
Category Lookup
Category
Furniture
Clothing
Electronics
To table
Order Details
Amount
Category
Order ID
Profit
Quantity
Sub-Category
561
Clothing
B-25602
212
3
Saree
119
Clothing
B-25602
-5
8
Saree
193
Clothing
B-25603
-166
3
Saree
Cardinality
One to many (1:*)
Cross-filter direction
Single
Make this relationship active
Apply security filter in both directions
Assume referential integrity
Save
Cancel
Visualizations
Build visual
Filters
Data
Search
Average Profit by
Category Lookup
List of Orders
Merged Orders Di
Order Details
Order ID Count
Sales target
Total Amount by S
Total Target by Mo
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here

New relationship

Select tables and columns that are related.

From table

List of Orders

City	CustomerName	Order Date	Order ID	State
Indore	Hitika	23 April 2018	B-25627	Madhya Prad...
Indore	Ashmi	26 April 2018	B-25637	Madhya Prad...
Indore	Yaanvi	01 May 2018	B-25645	Madhya Prad...

To table

Order Details

Amount	Category	Order ID	Profit	Quantity	Sub-Category
561	Clothing	B-25602	212	3	Saree
119	Clothing	B-25602	-5	8	Saree
193	Clothing	B-25603	-166	3	Saree

Cardinality

One to many (1:*)

Cross-filter direction

Single

- [x] Make this relationship active
- [ ] Apply security filter in both directions
- [ ] Assume referential integrity

Save Cancel